

Introduction

Legged robot designs are often inspired by animals in nature. However, these designs may not be optimized for task-specific performance, and only encapsulate a small fraction of the design space. Animals have developed distinct morphological characteristics that enable them to compete and survive in their environment. Can we automatically search the space of robot designs in a similar fashion? In this work, we propose an efficient evolutionary algorithm that designs a robot from scratch, given a task and a simulation environment.

Background

Morphology representations. Soft robots are usually represented with a 3D grid of voxels with varying physical properties. Rigid robot designs are represented with kinematic trees, which consist of “bodies” interacted with “joints”. Physics simulators such as *MuJoCo*, enable us to physically simulate robots from a kinematic tree. We adopt the latter formulation, following the approach from Sims [3].

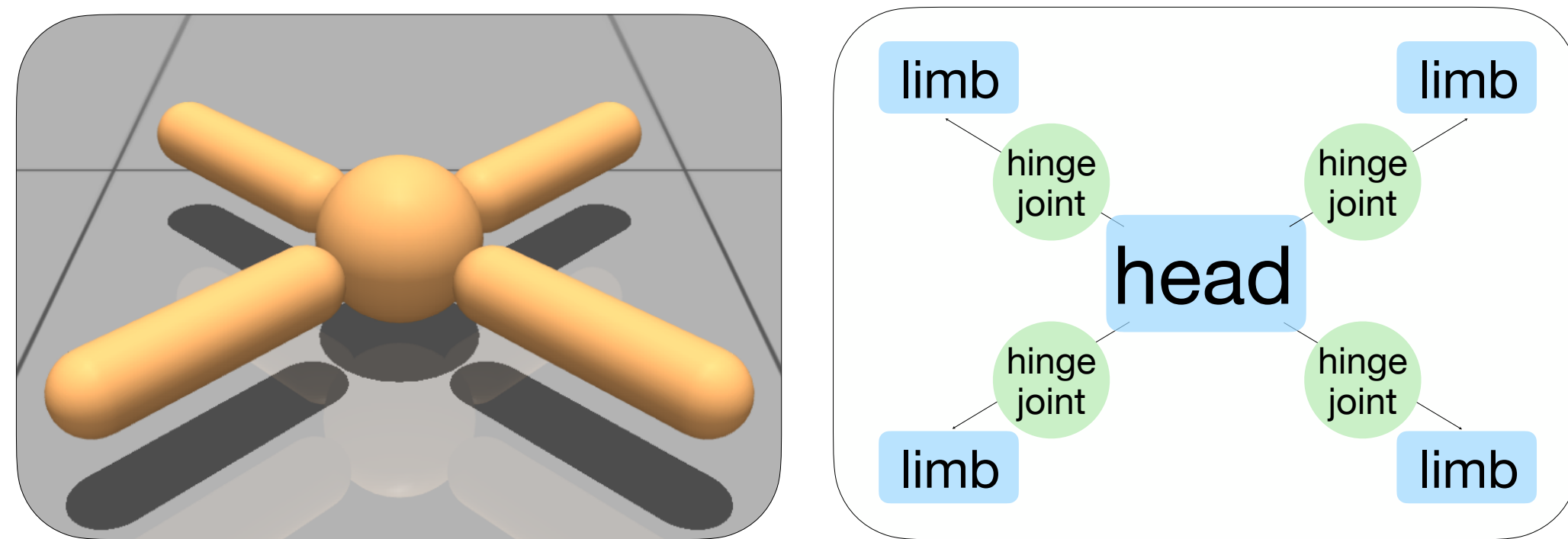


Figure 1: The kinematic tree describing the “MuJoCo Ant”.

Design agnostic control. Because legged robot designs can be very different structurally, a unique controller is required to score each morphology to its true ability. Thus, robot evolution, has so far required the co-design of morphologies and their respective controllers. We formulate the control problem as a Markov decision process, pictured below.

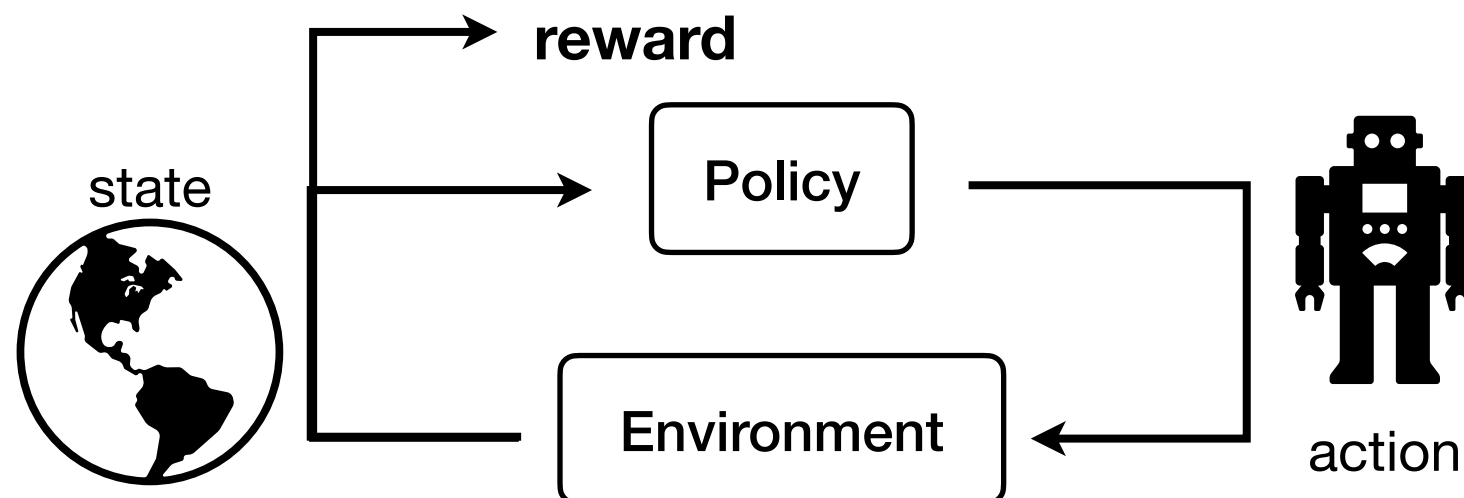


Figure 2: A Markov decision process consists of an environment, which outputs a state and reward, and a policy which outputs an action.

Reinforcement learning optimizes a controller which maps from a state to a policy. Due to their stability, policy gradient methods such as PPO have been used in prior works to evaluate each candidate in the evolutionary process [1]. However, it typically requires billions of simulation steps per morphology, making it unsustainable for large population sizes and generation counts.

Model predictive control optimizes a policy online, during the controller rollout, by periodically updating an action plan based on “imagined” rollouts. While it doesn’t result in a reusable controller at the end of optimization, it enables sample efficient evaluation of candidate robot morphologies in specific scenarios.

Hypothesis

We hypothesize that a general physics simulator, combined with model predictive control can enable efficient comparison and discovery of robot designs based on task success.

Rapid discovery of robot designs via model predictive control

Mihika Kamat

Method

We begin with a population of 128 procedurally generated morphologies. Each morphology starts with a *head body*, and receives a randomly chosen sequence of mutations. At each generation, every member of the population is evaluated for fitness. The best performing morphology per group of 4 is mutated and used to replace the oldest members of the population. Further details are outlined in *Algorithm 1*.

Algorithm 1 Evolutionary Predictive Sampling (EPS)

Require: initial population $\mathcal{P}$, objective J

Evolve: (repeat)

1. **Evaluate** each design in population $\mathcal{P}$ for task fitness
2. Randomize groups of size k
3. Mutate best performing design in each group and insert into $\mathcal{P}$
4. Delete $|\mathcal{P}|/k$ oldest members in population

Evaluate: (repeat)

1. **Plan** to update policy $\Pi(\theta)$
2. Read the current action u from policy $\Pi(\theta)$, apply it to the controlled system
3. Accumulate J to compute fitness

Plan: (repeat)

1. Measure the current state x
2. Get $N-1$ samples $\theta^{(i)} \sim \mathcal{N}(\theta, \sigma^2)$
3. Including θ , roll out all N samples from x
4. Pick the best one: $\theta \leftarrow \operatorname{argmax}(J(\theta^{(i)}))$

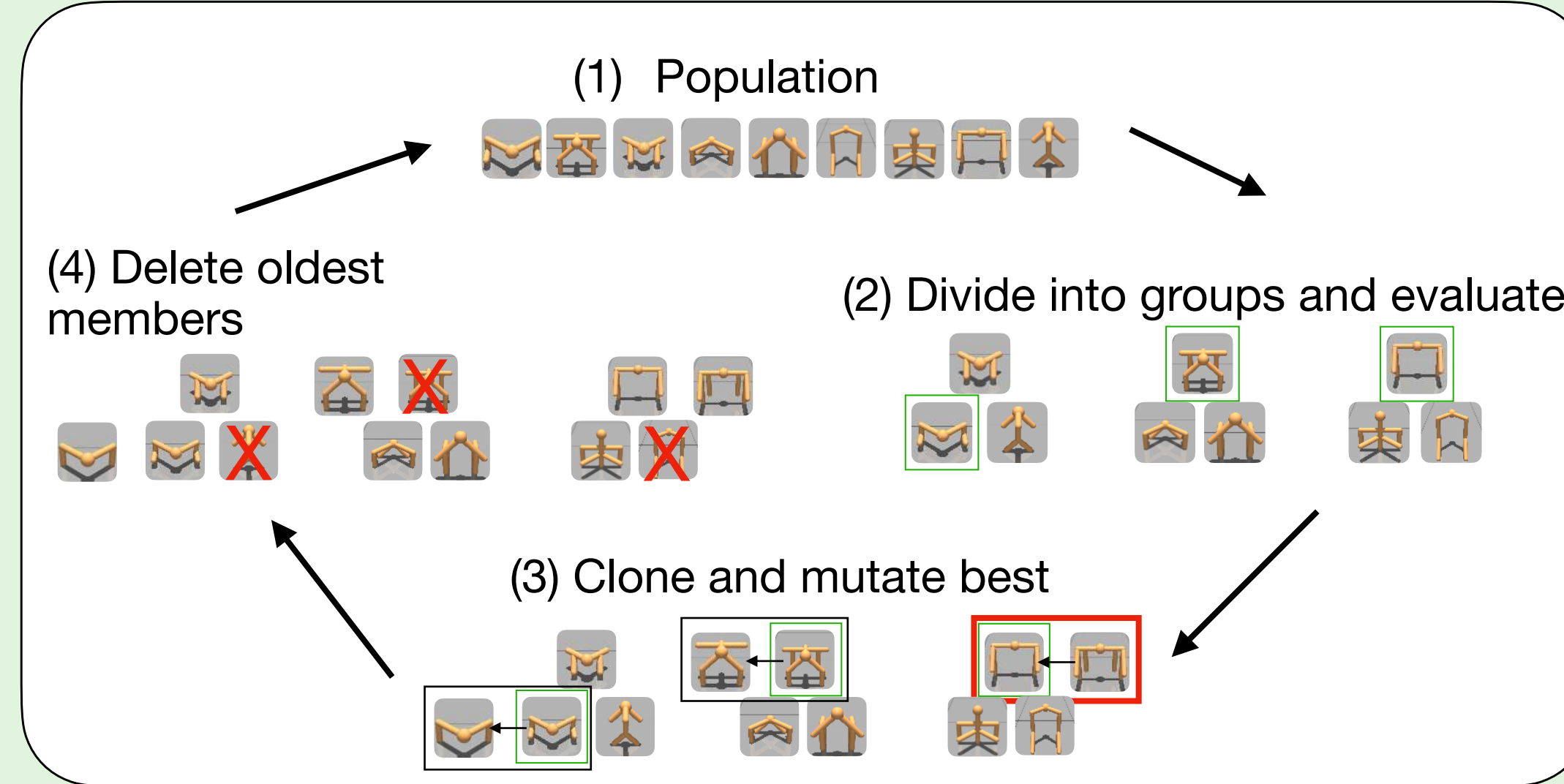


Figure 3: Evolution cycle on sample population.

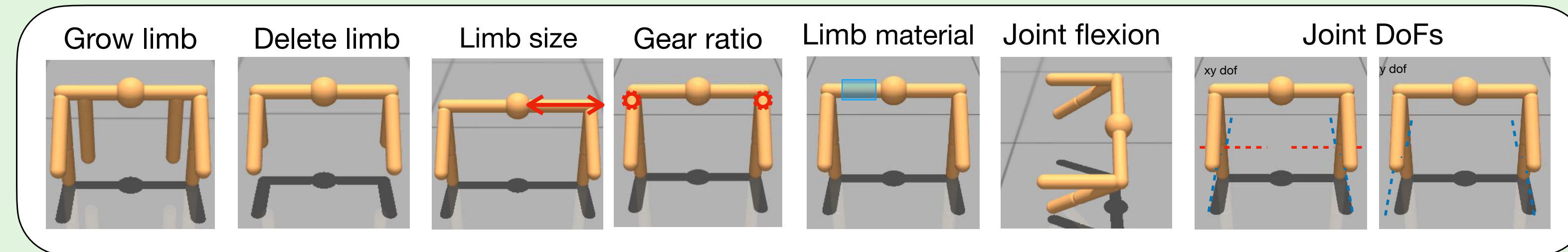
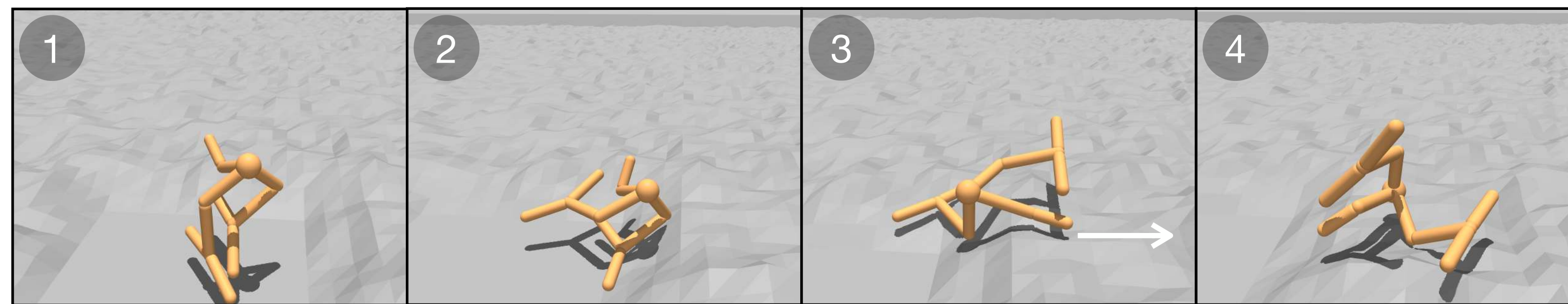


Figure 4: List of mutation primitives used for initializing and modifying designs.

Experiments

To test our hypothesis, we design four tasks: rough terrain locomotion, flat terrain locomotion, transporting a cube to a goal, and reorienting a cube to a goal orientation. Their individual performance as reward, with key components outlined below.

a) Rough terrain locomotion

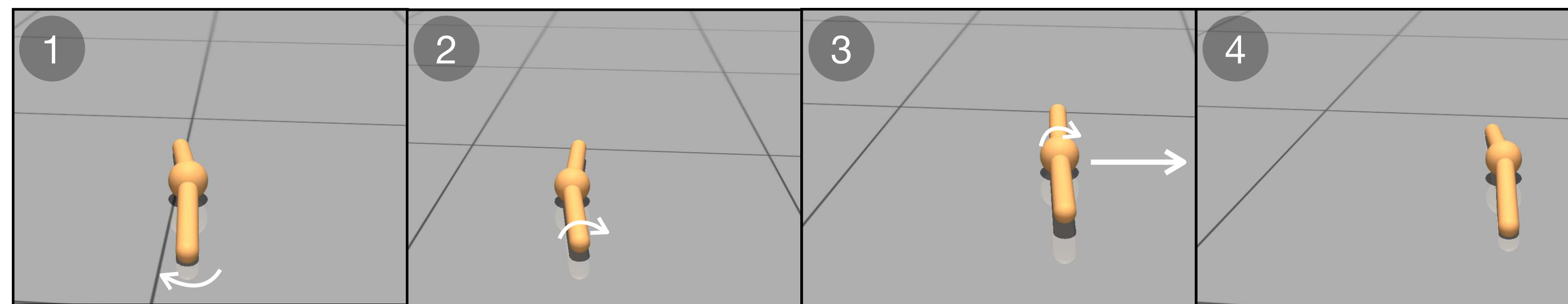


Reward function

$$J = \exp\left(\frac{-(v - v^*)^2}{\sigma^2}\right)$$

Target velocity $v^* = 1$ m/s.

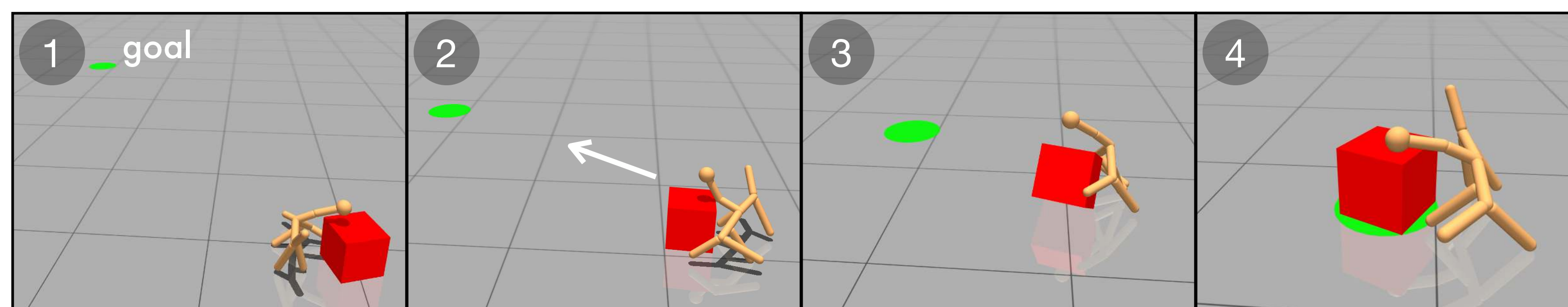
b) Flat terrain locomotion



$$J = \exp\left(\frac{-(v - v^*)^2}{\sigma^2}\right)$$

Target velocity $v^* = 1$ m/s.

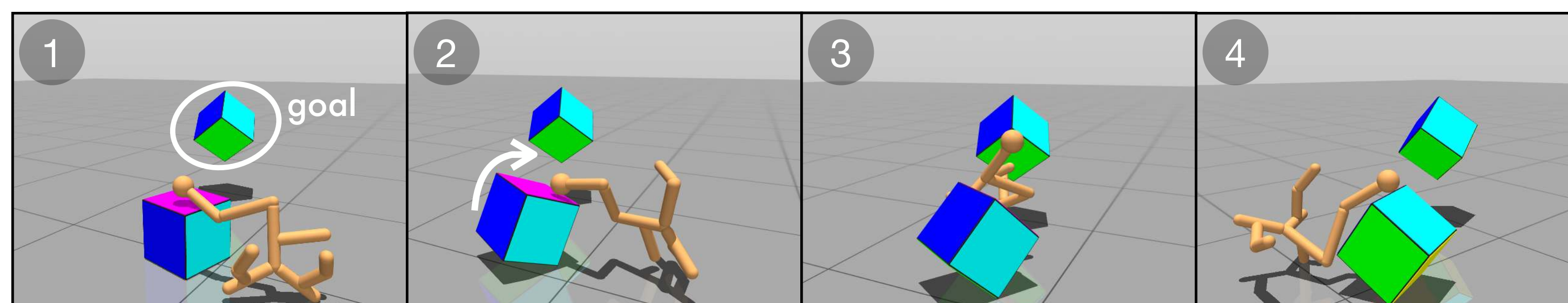
c) Transport cube to goal



$$J = \log\left(\frac{d_{\text{initial}} + \epsilon}{d_{\text{current}} + \epsilon}\right) + R_{\text{shaping}}$$

Goal sampled in 25m x 25m grid.

d) Reorient cube to goal



$$J = \exp\left(\frac{-\operatorname{acos}(|q \cdot q^*|)}{\sigma^2}\right) + R_{\text{shaping}}$$

Goal orientation $q^* \sim S^3$.

Results

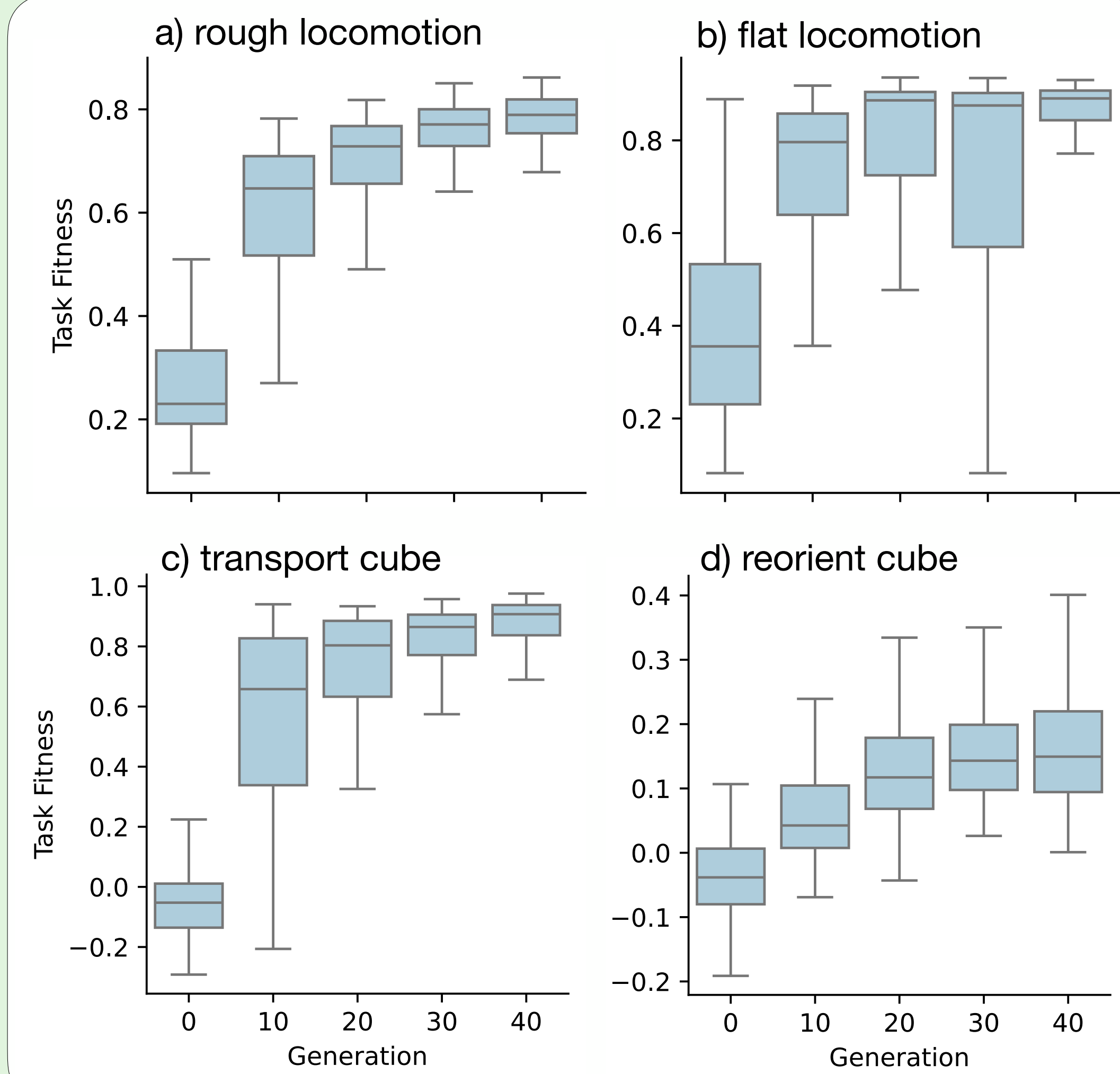


Figure 5: Distribution of fitness score for morphologies versus evolution generations completed for each task.

Over 50 generations, design performance improves. However, diversity in designs reduces in terms of skeletal structure, but continues to exhibit variance in limb length.

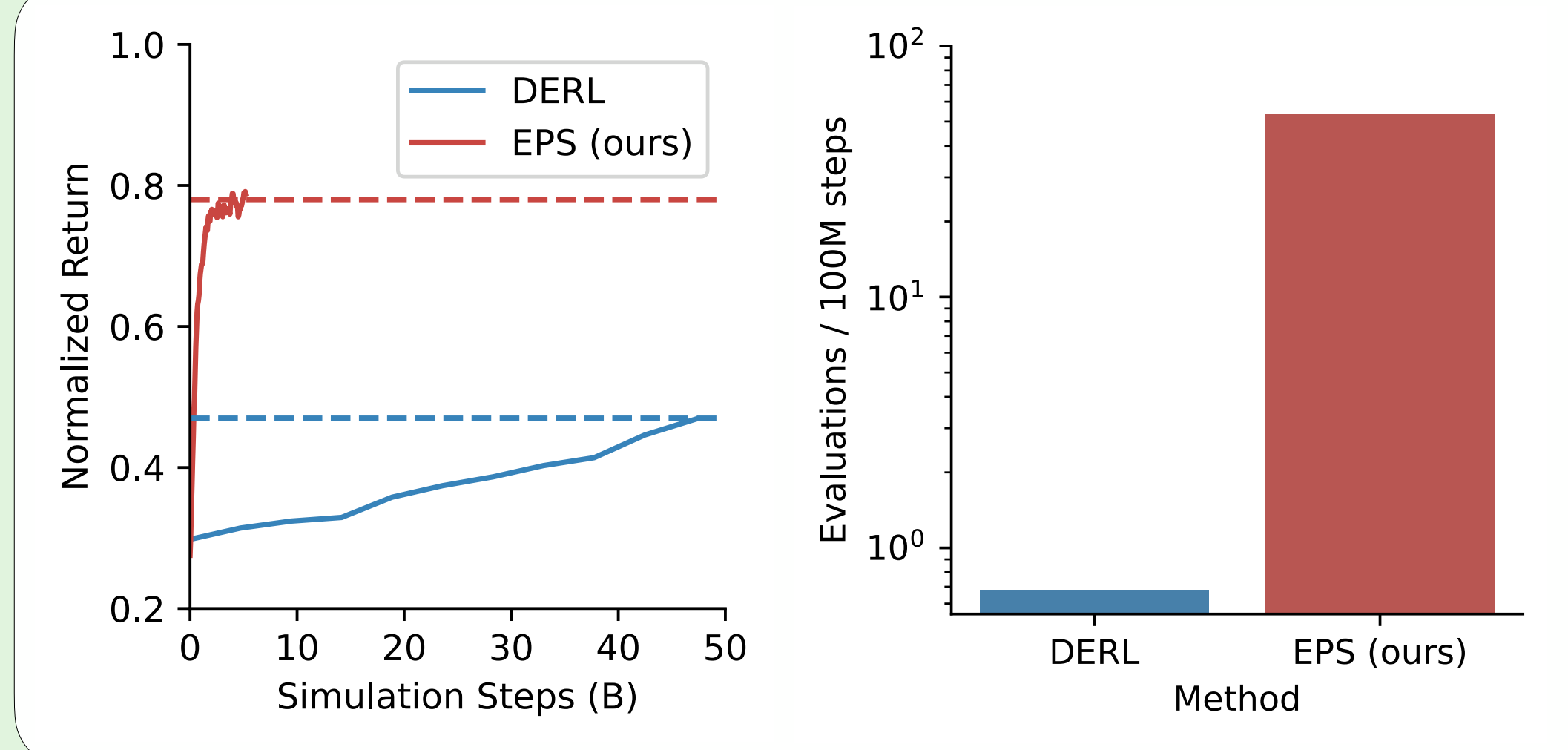


Figure 6: Comparing EPS to DERL (an RL baseline)

EPS achieves 80x more evaluation throughput than DERL as measured in simulation steps. In terms of improvement per generation, these two methods are comparable.

Future Work

Overcoming sparse rewards. Predictive sampling relies on dense rewards, due to a finite planning horizon. While this method works well for locomotion tasks, it is more challenging to reward manipulation tasks. Future work could explore the use of value functions to overcome these limitations.

Hybrid Tasks. To make EPS useful for real world robot design, morphologies could be evaluated on multiple tasks simultaneously.

References

1. Gupta, A., Savarese, S., Ganguli, S., et al. (2021). Embodied intelligence via learning and evolution. *Nature Communications*, 12, 5721.
2. Howell, T., Gileadi, N., Tunyasuvunakool, S., Zakka, K., Erez, T., & Tassa, Y. (2022). “Predictive Sampling: Real-time Behaviour Synthesis with MuJoCo,” arXiv:2212.00541.
3. Sims, K. (1994). Evolving virtual creatures. In *Proceedings of the 21st Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '94)*. Association for Computing Machinery.

Special thanks to Adam Rashid for his mentorship.